Contribution Results

Reliable Hybrid SLC/MLC ReRAM Crossbars for Transformers

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Step 1: What is compared

The baseline is the HyFlexPIM hybrid (SVD components ranked by the loss gradient, 20% in SLC) with the all-SLC and all-MLC chips as references; it is documented in the separate *Baseline Results*. Our four contributions are evaluated with the same cell-level simulator on BERT-Base and DistilBERT (SST-2) and, for contributions 1 and 2, on MNLI:

1. **Magnitude placement**: rank components by singular value $|\sigma_r|$ instead of the gradient.
2. **Fault-aware mapping**: after a chip is tested, place each component in the slot whose stuck cells hurt it least (Fisher-weighted Hungarian assignment), using the free order of SVD components.
3. **Tier-aware training**: fine-tune on a freshly simulated chip at every step, each component under the errors of the tier (SLC or MLC) it will be stored in.
4. **Generalisation to MNLI** of contributions 1 and 2.

All placements, mappings and trained models are evaluated on the same simulated chips (same seeds), so differences within an experiment are paired and given as mean ± standard error over chips. The hybrid chip uses 9.2 cells per weight, 34% fewer than all-SLC (14).

Step 2: The baseline's gaps

The baseline document shows three gaps, which the contributions address in turn: (i) the HyFlexPIM gradient placement falls behind all-SLC at high programming noise; (ii) stuck cells break every baseline, all-MLC at 0.1% and the HyFlexPIM hybrid near chance at 1%; (iii) even all-SLC loses 14–27 points at 1% stuck cells.

Step 3: Contribution 1, magnitude placement

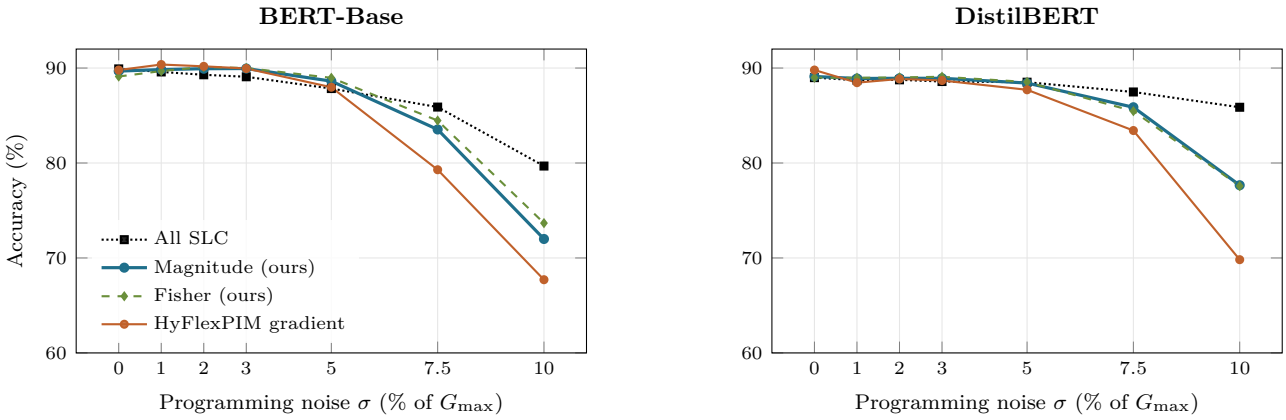


Figure 1: Accuracy against programming noise, 20% SLC, mean of 5 chips.

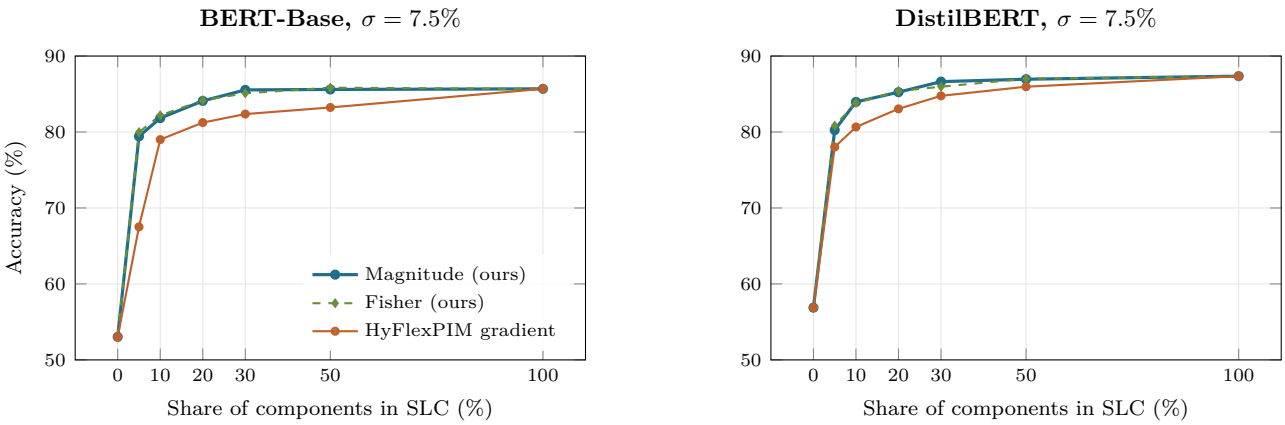


Figure 2: Accuracy against the share of components in SLC at $\sigma = 7.5\%$, mean of 20 chips; 0% is all-MLC, 100% all-SLC.

Table 1: Paired difference to magnitude, in correctly classified sentences out of 872 (mean ± standard error; 100 chips per noise level, 300 overall, all SLC shares).

Model	Placement	$\sigma = 3\%$	5%	7.5%	All
BERT-Base	Fisher	-0.1 ± 0.5	-0.1 ± 0.7	$+1.4 \pm 1.9$	$+0.4 \pm 0.7$
	HyFlexPIM gradient	-1.2 ± 0.7	-10.4 ± 1.6	-40.4 ± 5.0	-17.3 ± 2.0
DistilBERT	Fisher	-0.4 ± 0.7	-1.3 ± 0.7	$+0.0 \pm 1.7$	-0.6 ± 0.7
	HyFlexPIM gradient	-0.9 ± 0.7	-4.0 ± 1.0	-18.4 ± 2.6	-7.8 ± 1.1

Table 2: Smallest SLC share within half a point of all-SLC (shares tested: 5, 10, 20, 30, 50%), and the resulting cells per weight relative to all-SLC.

Model	Placement	$\sigma = 3\%$	$\sigma = 5\%$	$\sigma = 7.5\%$
BERT-Base	Magnitude (ours)	**5%** (59% of cells)	**5%** (59%)	**30%** (70%)
	HyFlexPIM gradient	5% (59%)	10% (61%)	not reached at 50%
DistilBERT	Magnitude (ours)	5% (59%)	**10%** (61%)	**50%** (79%)
	HyFlexPIM gradient	5% (59%)	20% (66%)	not reached at 50%

Analysis.

- Magnitude and Fisher are statistically indistinguishable at every noise level, while the HyFlexPIM gradient falls further behind as noise grows (Table 1); at 5% SLC and $\sigma = 7.5\%$ the gap on BERT-Base is 11.9 ± 1.7 points.
- **Why.** Each crossbar column has its own 8-bit scale, and the column of component r holds $\sigma_r v_r$, so the weight error a fixed cell noise causes is proportional to $|\sigma_r|$. The Fisher score multiplies this error size by the loss sensitivity; that it selects the same components as magnitude suggests the error size dominates. The gradient measures sensitivity alone and misses it.
- **Cost.** Magnitude needs half the SLC share of the gradient at $\sigma = 5\%$ and is the only score that reaches all-SLC at $\sigma = 7.5\%$ (Table 2). It needs no data or gradients.

Step 4: Contribution 2, fault-aware mapping

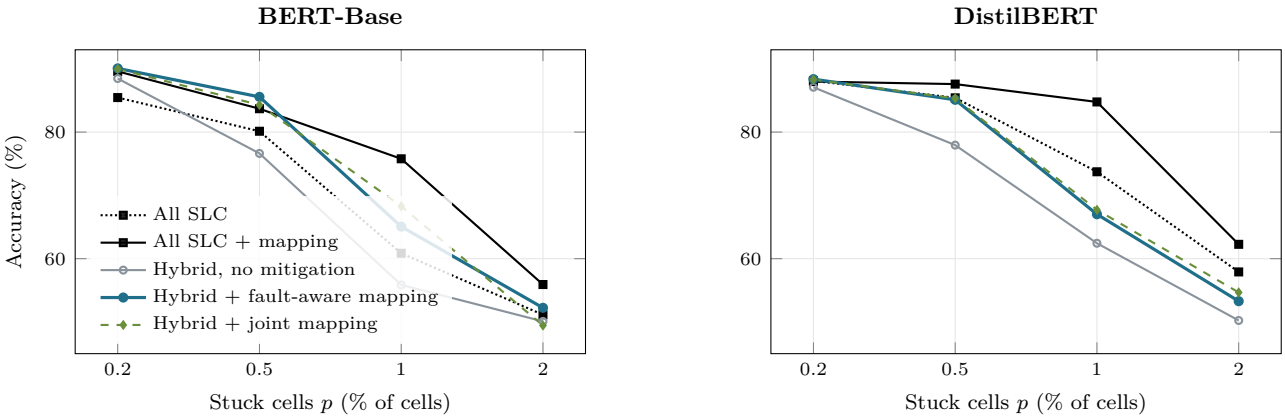


Figure 3: Accuracy against the share of stuck cells, no programming noise, mean of 10 chips. The hybrid uses magnitude placement with 20% SLC.

Table 3: Paired gain of fault-aware mapping over no mitigation (hybrid, magnitude, 20% SLC), and of joint tier-and-slot mapping over fault-aware mapping, in accuracy points (mean ± s.e. over 10 chips).

		Stuck cells, no noise				Stuck cells, $\sigma = 5\%$			1% stuck	
	Model	0.2%	0.5%	1%	2%	0.2%	0.5%	1%	clusters	lines
Mapping gain	BERT	$+1.6 \pm 0.3$	$+8.9 \pm 2.0$	$+9.2 \pm 2.6$	$+2.1 \pm 1.2$	$+2.5 \pm 0.6$	$+10.2 \pm 2.1$	$+6.0 \pm 1.7$	$+10.6 \pm 2.3$	$+6.8 \pm 2.2$
	Distil	$+1.3 \pm 0.5$	$+7.2 \pm 3.3$	$+4.6 \pm 3.4$	$+3.1 \pm 1.1$	$+2.3 \pm 0.5$	$+8.1 \pm 1.2$	$+5.6 \pm 3.5$	$+12.6 \pm 2.1$	$+13.9 \pm 3.3$
Joint − mapping	BERT	-0.1 ± 0.2	-1.3 ± 0.8	$+3.2 \pm 2.2$	-2.8 ± 0.9	-0.3 ± 0.3	-2.5 ± 1.9	$+3.6 \pm 2.0$	-2.8 ± 2.7	-0.1 ± 1.9
	Distil	-0.1 ± 0.1	$+0.2 \pm 0.7$	$+0.7 \pm 2.0$	$+1.4 \pm 1.6$	-0.7 ± 0.4	-0.7 ± 0.8	-2.0 ± 3.1	$+7.0 \pm 2.3$	-5.3 ± 2.4

Analysis.

- **Mapping helps in every setting** (Table 3), with or without programming noise. Gains are largest for clustered and line faults, where moving important components away from a damaged region matters most, and chips become more alike: at 0.5% stuck cells the standard deviation over BERT-Base chips falls from 6.7 to 1.2 points.
- **Choosing the tier per chip does not help**: joint mapping wins and loses by similar margins, mostly within two standard errors. Its second-order cost underrates the large error of a stuck high-order cell, so the benefit comes from the choice of slot.
- **Mapping alone is not enough above 0.5% stuck cells** (Fig. 3): the mapped hybrid falls behind all-SLC with mapping, because MLC cells hold two bits and the SLC tier offers few slots. Step 5 closes this gap.

Step 5: Contribution 3, tier-aware training

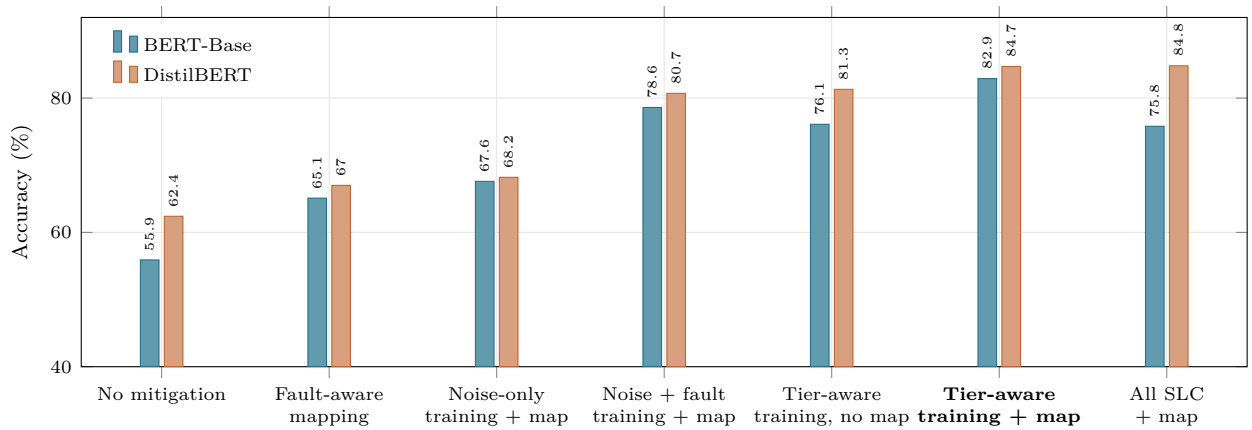


Figure 4: Accuracy at 1% stuck cells, no programming noise, mean of 10 chips. All bars except the last are the hybrid chip (magnitude placement, 20% SLC).

Table 4: Paired differences of tier-aware training + mapping, in accuracy points (mean ± s.e., 10 chips).

		Stuck cells, no noise				Stuck cells, $\sigma = 5\%$			1% stuck	
Compared with	Model	0.2%	0.5%	1%	2%	0.2%	0.5%	1%	clusters	lines
mapping only	BERT	-1.2 ± 0.2	$+2.7 \pm 0.5$	$+17.8 \pm 2.2$	$+0.8 \pm 1.0$	$+1.6 \pm 0.5$	$+7.4 \pm 0.8$	$+22.4 \pm 1.8$	$+20.7 \pm 1.6$	$+20.8 \pm 2.8$
	Distil	-0.1 ± 0.2	$+3.0 \pm 0.8$	$+17.7 \pm 2.7$	$+4.1 \pm 2.4$	$+0.5 \pm 0.4$	$+3.9 \pm 0.6$	$+14.7 \pm 2.4$	$+14.9 \pm 2.4$	$+16.5 \pm 3.1$
noise + fault training	BERT	-1.0 ± 0.2	-0.4 ± 0.4	$+4.3 \pm 2.2$	$+2.2 \pm 1.0$	-0.6 ± 0.4	$+0.4 \pm 0.3$	$+7.5 \pm 1.8$	$+6.2 \pm 1.7$	$+5.1 \pm 3.7$
	Distil	-0.4 ± 0.2	-0.1 ± 0.2	$+4.0 \pm 0.6$	$+2.4 \pm 1.7$	-0.5 ± 0.3	$+0.5 \pm 0.3$	$+2.3 \pm 1.6$	$+2.7 \pm 1.3$	$+2.8 \pm 0.8$
all SLC + mapping	BERT	-0.8 ± 0.3	$+4.6 \pm 0.7$	$+7.1 \pm 3.3$	-2.9 ± 2.2	$+2.9 \pm 1.4$	$+5.4 \pm 1.0$	$+4.0 \pm 1.7$	$+3.7 \pm 1.4$	$+0.2 \pm 3.0$
	Distil	$+0.3 \pm 0.2$	$+0.5 \pm 0.5$	-0.0 ± 0.9	-4.9 ± 3.1	$+0.7 \pm 0.3$	$+0.1 \pm 0.3$	-0.9 ± 2.4	-2.2 ± 1.2	$+3.0 \pm 1.9$

Table 5: Accuracy (%) under programming noise alone (no stuck cells), hybrid with 20% SLC by magnitude, mean of 10 chips; $\sigma = 0$ is an error-free chip.

	BERT-Base			DistilBERT		
Training	$\sigma = 0$	5%	7.5%	0	5%	7.5%
No training	89.7	88.7	84.2	89.1	88.2	85.6
Noise-only training	89.9	89.7	86.3	87.4	87.6	85.6
Noise + fault training	**90.9**	**89.9**	**88.8**	**89.9**	**89.3**	87.6
Tier-aware training	89.4	89.3	88.3	87.8	88.4	**88.0**

Analysis.

- **Stuck cells must be injected during training**: noise-only training barely moves the bars of Fig. 4, while adding stuck cells gives the largest single step.
- **Knowing the tiers helps where errors are dense**: tier-aware training beats tier-unaware training at 1% stuck cells, with programming noise and for clustered faults, ties at 0.5%, and is slightly behind at 0.2%, the cost of training on the harsher chip (Table 5, $\sigma = 0$).
- **Training and mapping are complementary**: training makes the weights tolerate errors, mapping keeps sensitive values away from known faults; tier-aware training alone already exceeds mapping alone, and mapping still adds on top.
- **The gap to all-SLC closes** (last block of Table 4): with tier-aware training and mapping the hybrid beats all-SLC with mapping on BERT-Base and matches it on DistilBERT up to 1% stuck cells, with 34% fewer cells. At 2% every configuration fails.
- Training also improves robustness to programming noise alone (Table 5).

Step 6: Contribution 4, generalisation to MNLI

Table 6: MNLI (three classes, chance 33.3%): paired results, in accuracy points (mean ± s.e.). Raw accuracies are in Appendix A.6.

	BERT-Base	DistilBERT
Fisher − magnitude, $\sigma = 5\%$, all SLC shares (100 chips)	-0.2 ± 0.2	-0.2 ± 0.1
HyFlexPIM gradient − magnitude, same chips	-3.4 ± 0.4	-0.5 ± 0.2
Fault-aware mapping gain at 0.5% stuck cells (10 chips)	$+20.1 \pm 3.0$	$+18.8 \pm 3.2$

Both contributions transfer to the harder task: magnitude again matches Fisher and beats the gradient on BERT-Base, and mapping recovers about 20 points at 0.5% stuck cells. MNLI also exposed a simulator limitation: the single 8-bit scale of the vector between a tier's two crossbars costs up to 11 points even on an error-free chip, so only placements and mappings with the same tier sizes are compared.

Step 7: Overall comparison with the baseline

Table 7: Accuracy (%) of the baseline and of our methods, BERT-Base / DistilBERT. All hybrids keep 20% of the components in SLC unless stated. Programming-noise rows use the same 20 chips for all columns; stuck-cell rows take the baseline from the stuck-cell and fault-pattern sweeps and our methods from the mapping experiment (10 chips), so those comparisons are not paired.

Scenario	HyFlexPIM baseline	Magnitude	+ fault-aware mapping	+ tier-aware training	All SLC + mapping
$\sigma = 7.5\%$, 20% SLC	81.2 / 83.1	**84.1 / 85.2**	–	–	85.7 / 87.4[†]
$\sigma = 7.5\%$, 5% SLC	67.5 / 78.0	**79.4 / 80.2**	–	–	85.7 / 87.4[†]
0.5% stuck cells	71.0 / 74.5	76.6 / 77.9	85.6 / 85.1	**88.3 / 88.1**	83.7 / 87.6
1% stuck cells	52.8 / 55.4	55.9 / 62.4	65.1 / 67.0	**82.9 / 84.7**	75.8 / 84.8
1% stuck, 3×3 clusters	51.5 / 56.4	53.7 / 55.9	64.3 / 68.5	**85.0 / 83.4**	81.3 / 85.6
1% stuck, bit lines	51.0 / 52.2	50.1 / 54.0	56.9 / 68.0	**77.7 / 84.4**	77.5 / 81.5

[†]all-SLC without mapping (mapping has no effect without stuck cells).

Findings.

- **Placement**: magnitude ranking matches the Fisher estimate of expected damage and beats the HyFlexPIM gradient at high noise, by $+2.8 \pm 0.8$ (BERT-Base) and $+2.2 \pm 0.3$ points (DistilBERT) at 20% SLC and $+11.9 \pm 1.7$ points at 5% SLC on BERT-Base, on the same chips.
- **Fault-aware mapping**: a zero-hardware reordering that improves the hybrid in every stuck-cell setting, by up to 13.9 points; choosing the tier per chip adds nothing.
- **Tier-aware training**: together with mapping, raises the hybrid at 1% stuck cells from 52.8% and 55.4% (HyFlexPIM baseline) to 82.9% and 84.7%, matching or beating an all-SLC chip with mapping up to 1% scattered stuck cells while using 34% fewer cells.
- **Generality**: the placement and mapping results hold on MNLI.

Appendix: complete results

Every experiment with every placement, both models and every sweep point, generated directly from the result files of the repository `github.com/Shivanshmittal023/reliable-reram-crossbar`. Each numbered step folder (`<step>`) stores its results as `<step>/results/<experiment>/<model>/sweep_summary.csv` (mean, standard deviation and worst chip per point), next to `sweep.csv` with one row per chip; the source folder of every section is given below. Accuracy in %, mean over chips; hybrids keep 20% of the components in SLC unless stated.

A.1 Programming noise (5 chips per point)

Source: `02_programming_noise/results/noise/`.

Table 8: Accuracy against programming noise σ.

Model	Placement	0%	1%	2%	3%	5%	7.5%	10%
BERT-Base	All SLC	89.9	89.6	89.3	89.1	87.8	85.9	79.7
	All MLC	90.0	89.3	87.3	83.9	66.3	52.2	51.3
	Random	90.0	90.1	89.9	87.7	78.8	58.8	49.1
	HyFlexPIM gradient	89.8	90.4	90.2	90.0	88.0	79.3	67.7
	Magnitude (ours)	89.7	89.8	89.9	90.0	88.6	83.5	72.0
	Fisher (ours)	89.1	89.7	90.1	90.0	89.0	84.5	73.7
DistilBERT	All SLC	89.0	88.7	88.8	88.6	88.5	87.5	85.9
	All MLC	89.1	88.4	87.9	86.1	74.5	60.4	56.0
	Random	88.6	88.7	88.7	87.8	82.8	61.8	52.4
	HyFlexPIM gradient	89.8	88.5	88.9	88.7	87.7	83.4	69.8
	Magnitude (ours)	89.1	88.9	88.9	88.9	88.4	85.9	77.7
	Fisher (ours)	89.0	89.0	89.0	89.1	88.5	85.5	77.6

A.2 SLC share

Source: `03_slc_share/results/slc_share/`.

Table 9: BERT-Base: accuracy by share of components in SLC (20 chips per point).

σ	Placement	0%	5%	10%	20%	30%	50%	100%
3%	Magnitude (ours)	82.1	89.7	89.9	89.8	89.8	89.7	89.2
	Fisher (ours)	82.1	89.6	89.8	89.8	89.8	89.8	89.2
	HyFlexPIM gradient	82.1	88.7	90.0	90.0	89.9	89.6	89.2
	Random	82.1	87.0	86.9	87.6	87.3	89.0	89.2
5%	Magnitude (ours)	66.8	88.0	88.6	88.7	88.9	88.9	88.0
	Fisher (ours)	66.8	87.9	88.6	88.8	88.9	88.9	88.0
	HyFlexPIM gradient	66.8	84.2	88.2	88.4	88.4	88.1	88.0
	Random	66.8	72.5	73.6	76.2	77.8	83.4	88.0
7.5%	Magnitude (ours)	53.0	79.4	81.8	84.1	85.5	85.6	85.7
	Fisher (ours)	53.0	80.0	82.2	84.2	85.1	85.8	85.7
	HyFlexPIM gradient	53.0	67.5	79.0	81.2	82.4	83.2	85.7
	Random	53.0	53.6	54.2	55.2	57.5	67.3	85.7

Table 10: DistilBERT: accuracy by share of components in SLC (20 chips per point).

σ	Placement	0%	5%	10%	20%	30%	50%	100%
3%	Magnitude (ours)	86.0	88.4	88.6	88.8	88.9	88.7	88.7
	Fisher (ours)	86.0	88.5	88.7	88.7	88.8	88.6	88.7
	HyFlexPIM gradient	86.0	88.4	88.5	88.6	88.8	88.7	88.7
	Random	86.0	86.4	86.8	87.5	88.1	88.4	88.7
5%	Magnitude (ours)	74.8	87.4	88.0	88.1	88.8	88.4	88.2
	Fisher (ours)	74.8	87.3	87.8	88.4	88.1	88.3	88.2
	HyFlexPIM gradient	74.8	87.3	87.3	87.7	88.1	87.8	88.2
	Random	74.8	75.4	77.4	81.9	82.3	85.7	88.2
7.5%	Magnitude (ours)	56.9	80.2	84.0	85.2	86.6	86.9	87.4
	Fisher (ours)	56.9	80.8	83.8	85.4	86.0	87.0	87.4
	HyFlexPIM gradient	56.9	78.0	80.7	83.1	84.8	86.0	87.4
	Random	56.9	56.4	58.2	63.1	64.7	73.7	87.4

A.3 Stuck cells (20 chips per point, no noise)

Source: `04_stuck_cells/results/fault/`.

Table 11: Accuracy against the share of stuck cells p.

Model	Placement	0%	0.1%	0.2%	0.5%	1%	2%	5%
BERT-Base	All SLC	89.9	88.3	85.6	79.0	62.7	51.4	49.6
	All MLC	90.0	58.6	52.6	50.8	51.2	49.6	49.6
	Random	90.0	67.9	55.8	50.7	50.6	50.2	49.9
	HyFlexPIM gradient	89.8	89.4	87.5	71.0	52.8	50.2	49.5
	Magnitude (ours)	89.7	89.6	88.6	76.8	57.9	51.0	50.0
	Fisher (ours)	89.1	89.7	88.4	75.6	55.5	49.5	50.3
DistilBERT	All SLC	89.0	88.2	87.8	85.2	74.6	58.5	51.5
	All MLC	89.1	65.5	58.2	52.3	51.1	50.2	49.4
	Random	88.6	76.9	63.3	51.3	51.0	49.4	50.1
	HyFlexPIM gradient	89.8	88.1	86.7	74.5	55.4	51.0	50.5
	Magnitude (ours)	89.1	88.3	87.4	80.3	62.1	51.4	49.9
	Fisher (ours)	89.0	88.3	87.2	80.3	62.1	51.3	50.4

A.4 Other non-idealities

Source: `05_other_non_idealities/results/{read,level,cluster,irdrop,adc,adc_noise}/`.

Table 12: Read noise σ_r (5 chips).

Model	Placement	0%	0.5%	1%	2%	3%	5%
BERT-Base	All SLC	89.7	89.8	89.8	89.0	87.2	80.0
	All MLC	90.4	89.4	87.7	77.3	63.8	53.0
	HyFlexPIM gradient	90.3	90.5	90.3	89.8	88.3	76.6
	Magnitude (ours)	90.6	90.1	89.9	90.1	89.3	82.5
	Fisher (ours)	89.6	90.0	89.9	89.7	89.0	82.7
DistilBERT	All SLC	88.5	89.0	88.2	88.3	87.5	84.5
	All MLC	89.0	88.6	87.9	83.8	74.5	57.7
	HyFlexPIM gradient	88.6	88.9	88.5	88.1	86.8	80.5
	Magnitude (ours)	88.8	88.9	88.7	88.4	87.7	83.3
	Fisher (ours)	89.1	88.8	88.7	88.2	87.5	83.2

Table 13: MLC level-dependent noise at $\sigma = 5\%$ (10 chips).

Model	Placement	uniform	mid $\times 1.5$	mid $\times 2$
BERT-Base	All SLC	88.2	88.2	88.2
	All MLC	67.3	64.7	62.5
	HyFlexPIM gradient	88.3	88.1	87.8
	Magnitude (ours)	88.9	88.7	88.4
	Fisher (ours)	89.0	88.9	88.4
DistilBERT	All SLC	88.4	88.4	88.4
	All MLC	75.7	73.5	70.3
	HyFlexPIM gradient	88.0	87.5	87.1
	Magnitude (ours)	88.4	88.3	88.0
	Fisher (ours)	88.1	88.0	87.9

Table 14: Fault patterns at 1% stuck cells (10 chips).

Model	Placement	scattered	clusters	lines
BERT-Base	All SLC	61.3	66.7	57.0
	All MLC	50.7	50.8	50.4
	HyFlexPIM gradient	52.6	51.5	51.0
	Magnitude (ours)	56.4	54.0	50.4
	Fisher (ours)	55.6	53.1	52.2
DistilBERT	All SLC	74.2	80.3	71.2
	All MLC	51.2	51.2	50.8
	HyFlexPIM gradient	55.0	56.4	52.2
	Magnitude (ours)	62.5	56.1	54.7
	Fisher (ours)	60.0	57.3	56.3

Table 15: IR drop, wire resistance r (deterministic).

Model	Placement	0	10^{-4}	3×10^{-4}	10^{-3}	3×10^{-3}
BERT-Base	All SLC	89.7	90.3	88.9	87.0	58.5
	All MLC	90.4	89.3	89.8	90.6	79.1
	HyFlexPIM gradient	90.3	90.5	90.5	88.6	76.4
	Magnitude (ours)	90.6	90.1	90.4	88.1	76.8
	Fisher (ours)	89.6	91.1	91.1	88.3	72.8
DistilBERT	All SLC	88.5	89.9	89.4	85.3	70.1
	All MLC	89.0	88.6	89.1	89.9	77.9
	HyFlexPIM gradient	88.6	89.6	89.6	86.9	74.7
	Magnitude (ours)	88.8	89.2	89.7	86.6	73.4
	Fisher (ours)	89.1	89.2	89.3	86.4	72.6

Table 16: Fixed ADC resolution with bit-serial inputs (deterministic).

Model	Placement	ideal	8 bit	7 bit	6 bit	5 bit	4 bit	3 bit
BERT-Base	All SLC	89.7	89.9	90.0	89.3	51.1	61.1	51.1
	All MLC	90.4	89.6	89.2	83.0	51.1	60.3	50.9
	HyFlexPIM gradient	90.3	90.1	90.1	89.6	53.0	63.2	51.9
	Magnitude (ours)	90.6	90.4	89.3	90.0	51.1	59.9	52.6
	Fisher (ours)	89.6	90.1	89.7	90.5	51.6	57.0	52.9
DistilBERT	All SLC	88.5	89.2	88.4	88.0	71.0	52.1	59.1
	All MLC	89.0	88.3	87.8	81.7	56.5	51.9	56.8
	HyFlexPIM gradient	88.6	88.4	88.5	88.4	77.3	51.5	60.1
	Magnitude (ours)	88.8	88.6	89.0	89.7	70.6	51.4	62.4
	Fisher (ours)	89.1	88.0	88.2	88.8	70.9	51.7	60.3

Table 17: Fixed ADC with $\sigma = 5\%$ programming noise (3 chips).

Model	Placement	ideal	8 bit	6 bit	4 bit
DistilBERT	All SLC	88.1	88.3	88.7	52.6
	All MLC	77.8	76.8	56.0	53.6
	HyFlexPIM gradient	87.8	87.6	87.2	51.8
	Magnitude (ours)	88.7	89.1	88.5	51.5
	Fisher (ours)	87.8	88.5	88.2	52.3

A.5 Fault-aware mapping (10 chips per point)

Source: `06_fault_aware_mapping/results/{fault,fault_noise,pattern}/`.

Table 18: Stuck cells, no noise.

Model	Placement	0.2%	0.5%	1%	2%
BERT-Base	All SLC	85.4	80.1	60.9	51.1
	All SLC + fault-aware mapping	89.6	83.7	75.8	55.9
	Hybrid, no mitigation	88.5	76.6	55.9	50.1
	Hybrid + fault-aware mapping	90.1	85.6	65.1	52.2
	Hybrid + joint mapping	90.0	84.3	68.3	49.4
DistilBERT	All SLC	88.1	85.4	73.7	57.9
	All SLC + fault-aware mapping	88.0	87.6	84.8	62.3
	Hybrid, no mitigation	87.1	77.9	62.4	50.2
	Hybrid + fault-aware mapping	88.4	85.1	67.0	53.3
	Hybrid + joint mapping	88.3	85.3	67.7	54.7

Table 19: Stuck cells with $\sigma = 5\%$.

Model	Placement	0.2%	0.5%	1%
BERT-Base	All SLC	85.6	78.4	64.6
	All SLC + fault-aware mapping	86.1	82.4	76.5
	Hybrid, no mitigation	84.9	70.3	52.1
	Hybrid + fault-aware mapping	87.4	80.4	58.1
	Hybrid + joint mapping	87.2	77.9	61.7
DistilBERT	All SLC	86.9	81.7	68.0
	All SLC + fault-aware mapping	87.6	86.8	80.7
	Hybrid, no mitigation	85.4	74.9	59.5
	Hybrid + fault-aware mapping	87.7	83.0	65.1
	Hybrid + joint mapping	87.0	82.3	63.1

Table 20: 1% stuck cells by pattern.

Model	Placement	clusters	lines
BERT-Base	All SLC	66.5	57.2
	All SLC + fault-aware mapping	81.3	77.5
	Hybrid, no mitigation	53.7	50.1
	Hybrid + fault-aware mapping	64.3	56.9
	Hybrid + joint mapping	61.5	56.8
DistilBERT	All SLC	79.6	71.1
	All SLC + fault-aware mapping	85.6	81.5
	Hybrid, no mitigation	55.9	54.0
	Hybrid + fault-aware mapping	68.5	68.0
	Hybrid + joint mapping	75.5	62.7

A.6 MNLI (2,000 matched validation pairs)

Source: `08_mnli/results/{slc_share,fault,mapping}/`.

Table 21: MNLI: accuracy by share of components in SLC at $\sigma = 5\%$ (20 chips per point).

Model	Placement	0%	5%	10%	20%	30%	50%	100%
BERT-Base	Magnitude (ours)	33.5	64.0	66.0	68.0	69.2	66.7	61.6
	Fisher (ours)	33.5	64.2	65.8	67.8	68.8	66.6	61.6
	HyFlexPIM gradient	33.5	59.6	63.5	63.3	66.5	64.2	61.6
	Random	33.5	34.1	33.9	35.9	38.9	44.9	61.6
DistilBERT	Magnitude (ours)	36.8	68.4	70.0	71.5	72.6	72.5	72.0
	Fisher (ours)	36.8	68.5	70.0	71.1	72.1	72.3	72.0
	HyFlexPIM gradient	36.8	66.6	69.0	71.3	72.1	73.6	72.0
	Random	36.8	38.3	39.3	44.1	48.1	55.3	72.0

Table 22: MNLI: stuck cells, 20% SLC (20 chips).

Model	Placement	0%	0.1%	0.2%	0.5%	1%	2%	5%
BERT-Base	All SLC	72.2	67.2	61.9	44.4	34.5	32.4	32.4
	All MLC	72.4	37.2	33.2	32.8	31.8	31.9	32.3
	Random	73.0	42.8	34.3	33.3	32.6	32.4	33.5
	HyFlexPIM gradient	78.2	73.0	65.2	36.9	32.0	31.5	32.4
	Magnitude (ours)	79.1	74.6	67.8	40.4	32.8	32.2	32.3
	Fisher (ours)	79.3	73.9	67.2	39.8	32.8	32.3	32.6
DistilBERT	All SLC	76.9	74.4	71.3	60.4	42.1	34.1	34.2
	All MLC	76.8	44.5	34.0	33.6	33.7	34.1	33.0
	Random	77.7	51.6	35.3	32.9	33.2	33.7	33.4
	HyFlexPIM gradient	79.6	76.0	71.4	44.5	33.9	32.9	33.9
	Magnitude (ours)	77.3	74.8	71.3	49.7	34.0	33.3	33.8
	Fisher (ours)	78.3	74.9	70.7	50.3	34.0	33.6	33.1

Table 23: MNLI: fault-aware mapping (10 chips).

Model	Placement	0.2%	0.5%	1%	2%
BERT-Base	All SLC	61.5	46.7	35.7	33.2
	All SLC + fault-aware mapping	69.9	61.9	46.2	33.1
	Hybrid, no mitigation	68.1	40.0	33.1	32.2
	Hybrid + fault-aware mapping	75.3	60.0	36.5	32.3
	Hybrid + joint mapping	75.5	60.4	35.5	32.3
DistilBERT	All SLC	71.4	58.4	41.4	34.6
	All SLC + fault-aware mapping	76.2	72.2	58.5	37.4
	Hybrid, no mitigation	69.9	49.1	33.4	33.4
	Hybrid + fault-aware mapping	76.6	67.9	38.8	34.3
	Hybrid + joint mapping	76.6	67.0	39.4	33.3

A.7 Training: every variant with and without fault-aware mapping (10 chips per point)

Source: `07_training/results/<variant>/<experiment>/`, with variants `nat_noise`, `nat_fault` and `tier_aware`, and experiments `fault`, `fault_noise` and `pattern`.

Table 24: Accuracy (%) of the hybrid chip (20% SLC by magnitude) for each training variant, without and with fault-aware mapping. The untrained model is in A.5.

			Stuck cells, no noise				Stuck cells, $\sigma = 5\%$			1% stuck	
Model	Training	Mapping	0.2%	0.5%	1%	2%	0.2%	0.5%	1%	clusters	lines
BERT-Base	Noise only	no	89.3	80.6	56.9	51.0	87.1	74.1	52.9	54.4	50.1
		yes	89.8	87.0	67.6	51.7	88.0	82.7	58.0	65.8	58.9
	Noise + faults	no	89.9	87.5	66.3	50.3	88.6	83.7	60.2	58.0	52.2
		yes	89.9	88.7	78.6	50.8	89.7	87.4	72.9	78.8	72.6
	Tier-aware	no	89.2	87.1	76.1	50.8	88.6	86.5	70.0	71.3	53.1
		yes	88.8	88.3	82.9	53.0	89.1	87.8	80.5	85.0	77.7
DistilBERT	Noise only	no	86.3	77.5	62.1	50.5	85.3	76.4	60.6	56.6	53.6
		yes	87.3	84.5	68.2	51.5	85.9	82.8	59.9	66.1	65.9
	Noise + faults	no	88.6	83.4	72.5	50.8	88.1	83.9	67.7	68.0	57.6
		yes	88.6	88.2	80.7	55.0	88.8	86.4	77.5	80.7	81.6
	Tier-aware	no	88.2	85.7	81.2	52.9	88.1	85.6	76.5	77.7	67.5
		yes	88.2	88.1	84.7	57.4	88.2	86.9	79.8	83.4	84.4

The training results under programming noise alone are in Table 5 (Step 5).